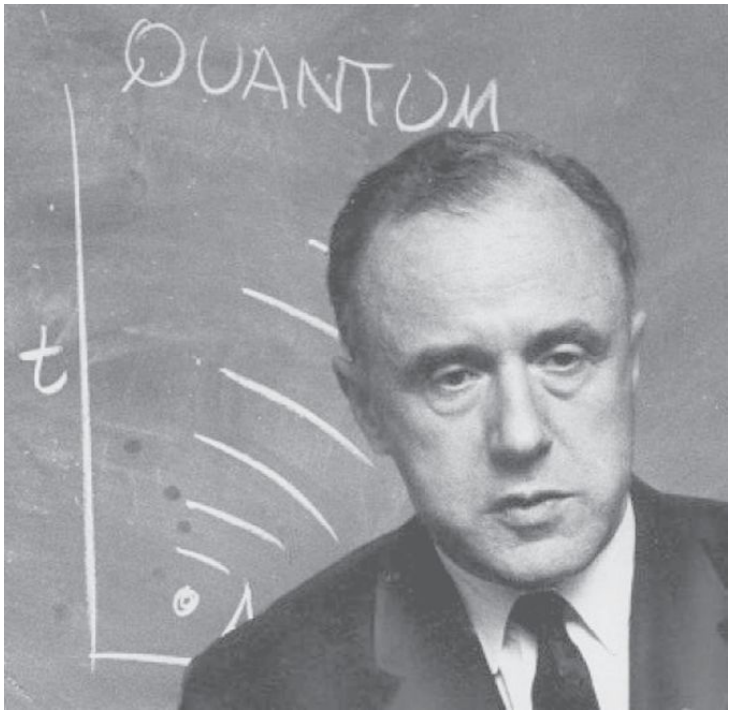


Figure 5.2 John Wheeler.

The scientist who has most contributed to quantum gravity is John Wheeler, a legendary figure who has traversed the physics of the past century. A pupil of and collaborator with Niels Bohr in Copenhagen; a collaborator with Einstein when Einstein moved to the United States; a teacher who can count among his students figures such as Richard Feynman … Wheeler was at the heart of the physics of the twentieth century. He was gifted with a fervid imagination. It was he who invented and made popular the term ‘black hole’. His name is associated with the early extended investigations – frequently more intuitive than mathematical – into how to think about quantum spacetime. Having absorbed Bronštejn’s lesson that quantum properties of the gravitational field imply a modification of the notion of space at a small scale, Wheeler looked for novel ideas to help conceive of this quantum space. He imagined it as a cloud of superimposed geometries, just as we can think of a quantum electron as a cloud of positions.

Imagine that you are looking at the sea from a great height: you perceive a vast expanse of it, a flat, cerulean table. Now you descend and look at it more closely. You begin to make out the great waves swollen by the wind. You descend further, and you see that the waves break up and that the surface of the sea is a turbulent frothing. This is what space is like, as imagined by Wheeler.[fn30] On our scale, immensely larger than the Planck length, space is smooth. If we move down to the Planck scale, it shatters and foams.

Wheeler sought a way to describe this foaming of space, this wave of probability of different geometries. In 1966, a young colleague of his who